EF101-P06: URL Shortener with Analytics

Course: EF-101 Fall 2026 **Application:** LinkLite **Group size:** 3 students

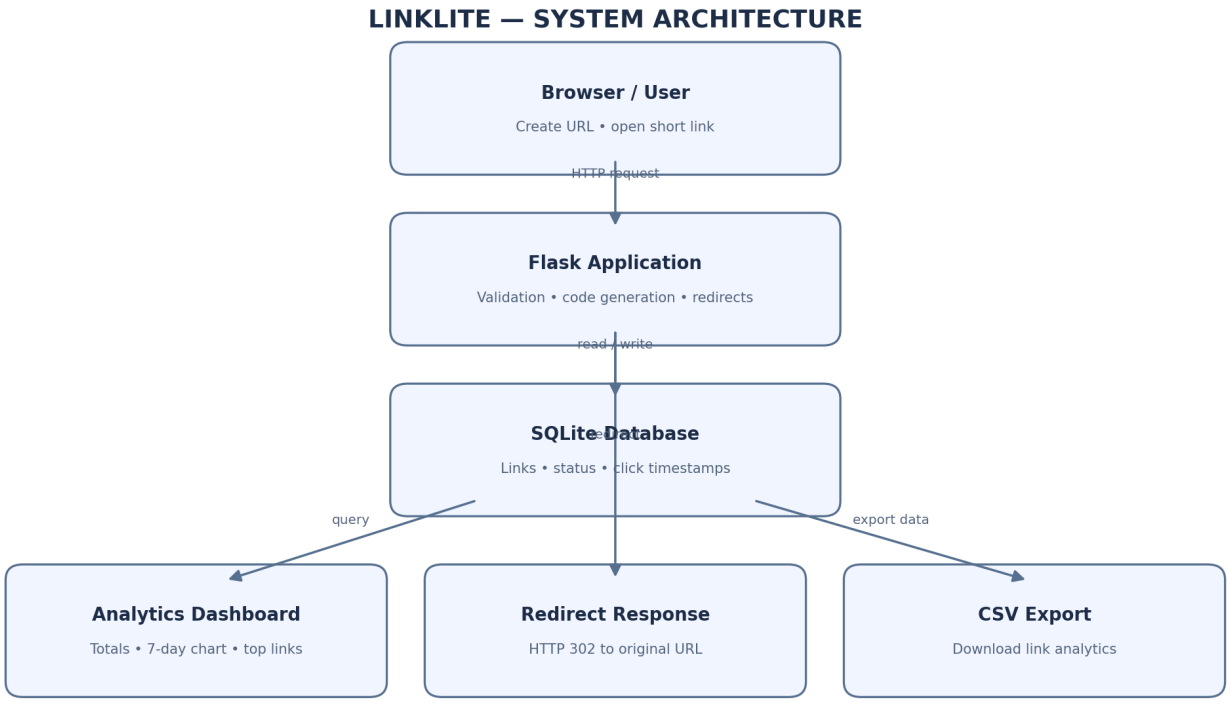
Objective

LinkLite accepts a long web address, generates a short URL, stores its mapping, redirects visitors to the original destination, and records click activity for basic analytics.

Main code segments

Code segment	Purpose
app.py — URL creation	Validates HTTP/HTTPS input, generates unique six-character codes, and stores links.
app.py — redirect route	Looks up codes, checks active status, records a timestamp per visit, and sends an HTTP 302 redirect.
SQLite storage	Persists links and click events across application restarts.
Link management routes	Search/filter is in the dashboard; links can be activated, deactivated, or deleted.
Analytics and CSV	Shows totals, seven-day clicks, top links and zero-click links; exports records to CSV.
templates/ and static/	Provide the web pages, tables, forms, responsive layout, and styling.
seed_demo.py	Optionally inserts 30 labelled demo links and 120+ simulated clicks for presentation.

System schema



Technology stack

Python 3, Flask, SQLite, HTML, CSS, and Python's CSV library. The application runs locally at http://127.0.0.1:5000.

Testing and responsible use

Demonstrate valid/invalid URLs, unique code creation, repeated clicks, persistence after restart, unknown and deactivated links, search/filter, CSV export, and analytics. Demo clicks are artificial and labelled as such. The application does not collect IP addresses and is intended for local classroom evaluation rather than unrestricted public deployment.

Repository name: EF101-P06-URL-Shortener-Analytics